

ISRU Water Extraction for Space Propellant: Monte Carlo Comparison of Asteroid and Lunar Sources at Industrial Scale

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Abstract

In-situ resource utilization (ISRU) for propellant production is widely considered essential for sustainable space infrastructure, yet no published model compares the major water sources—C-type near-Earth asteroids, lunar polar ice, and Phobos/Deimos—on a consistent economic basis accounting for extraction physics, transport costs, and parametric uncertainty. We present a Monte Carlo cost model that computes the net present value per kilogram of water delivered to the Earth–Sun L4/L5 Lagrange points from each source, propagating uncertainty in water fraction, extraction yield, energy cost, transport delta- v , and capital investment through 10,000 Monte Carlo trials.

Under baseline assumptions calibrated to OSIRIS-REx Bennu sample data (10% water fraction, 70% extraction yield) and current electric propulsion performance (2,500 s I_{sp}), C-type NEA water is delivered to L4/L5 at a median NPV cost of \$3,100/kg, compared to \$4,700/kg for lunar polar ice and \$16,300/kg for Phobos/Deimos. Monte Carlo analysis shows that NEA sources are cheaper than lunar sources in 91% of parameter draws. The cost advantage of asteroids is driven primarily by lower transport energy (electric propulsion from NEAs vs. chemical propulsion from the lunar surface) and higher water fraction (CI chondrite 5–15% vs. lunar PSR 2–10%). Lunar sources become competitive only when NEA delta- v exceeds ~ 6 km/s or

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¹This research was conducted using an AI-assisted multi-model consensus methodology described in [?]. Simulation code is open-source at <https://github.com/project-dyson>.

NEA water fraction falls below $\sim 4\%$, conditions that exclude most known C-type NEAs. We identify extraction yield and transport cost as the two parameters with the greatest influence on the crossover, and we propose a phased development strategy that begins with NEA water extraction while maintaining lunar capability as a hedge.

Keywords: ISRU, water extraction, asteroid mining, lunar ice, Monte Carlo, space propellant, near-Earth asteroid

1. Introduction

Water is the most versatile in-space resource. Electrolyzed into hydrogen and oxygen, it provides high-performance cryogenic propellant ($I_{sp} \approx 450$ s chemical, or feedstock for electric propulsion). Unprocessed, it serves as
5 radiation shielding, life support consumable, and thermal working fluid. For any large-scale space infrastructure program, the question is not whether to use ISRU water, but *where to get it*.

Three candidate sources dominate the literature: (1) C-type near-Earth asteroids, whose CI/CM chondrite compositions contain 5–15% water by
10 mass bound in hydrated phyllosilicates [?]; (2) lunar polar permanently shadowed regions (PSRs), where neutron spectrometry and direct measurement indicate 2–10% water ice in regolith [?]; and (3) the Martian moons Phobos and Deimos, whose spectral properties suggest possible hydration but whose water content is highly uncertain.

15 Previous economic analyses of ISRU water have been source-specific. Sowers [?] analyzed the business case for lunar ice mining, establishing break-even conditions against Earth-launched propellant. Sercel [?] developed the Optical Mining concept for asteroid volatile extraction, demonstrating full-scale thermal processing of CI-type simulant. Sanders and Larson [?]
20 provided NASA’s ISRU technology roadmap. None compares the sources on a consistent economic basis suitable for architectural decision-making.

This paper extends the parametric Monte Carlo framework of Paper 01 in this series [?] to the water extraction problem. We model each source’s capital costs, operating costs with learning curves, extraction physics, and
25 transport costs to L4/L5, then propagate parametric uncertainty through 10,000 Monte Carlo trials to produce probability distributions over cost per kilogram delivered. The key output is the *optimal source as a function of*

program scale and parameter uncertainty—the information needed by a program architect to make a robust source selection decision.

30 2. Water Source Characterization

2.1. *C-type near-Earth asteroids*

The OSIRIS-REx mission returned 121.6 g of material from C-type asteroid (101955) Bennu in September 2023. Preliminary examination [?] revealed a composition dominated by hydrated phyllosilicates (serpentine and
35 saponite), with magnetite, sulfides, carbonates, organic matter, and sodium-rich phosphates. The composition resembles the most chemically primitive CI chondrites.

Subsequent analyses have significantly enriched the volatile inventory:

- Glavin et al. [?] found Bennu samples more volatile-rich than Ryugu
40 samples, with higher carbon, nitrogen, and ammonia concentrations.
- McCoy et al. [?] identified 11 evaporite minerals formed from saltwater evaporation, providing direct evidence for long-duration liquid water on Bennu’s parent body.

For modeling purposes, we adopt a baseline water fraction of 10% (range
45 5–15%), consistent with CI chondrite bulk compositions and the Bennu sample results. The water is bound in phyllosilicate crystal structures and must be liberated by thermal processing at 400–700°C.

2.2. *Lunar polar ice*

The LCROSS impact experiment detected approximately 5.6% water ice
50 by mass in the Cabeus crater ejecta plume [?]. Subsequent orbital measurements from the Lunar Reconnaissance Orbiter suggest heterogeneous distribution, with some PSR locations potentially containing up to 10% water ice mixed with regolith, while other areas contain less than 2%.

Lunar water extraction faces distinct challenges: (1) operations in per-
55 manently shadowed craters require independent power sources, (2) the ice is likely mixed with regolith at depth rather than existing as pure ice deposits, and (3) transport to L4/L5 requires chemical propulsion for lunar ascent ($\Delta v \approx 2.4$ km/s) followed by transfer to the Lagrange point.

2.3. Phobos/Deimos

60 The Martian moons show spectral features possibly consistent with hy-
 drated minerals, but their actual water content is highly uncertain (0–5%).
 The JAXA MMX mission, expected to return samples from Phobos, may
 resolve this uncertainty. We include Phobos/Deimos as a third source to
 bracket the option space, but their high delta- v to L4/L5 (~ 6 km/s) makes
 65 them uncompetitive under most assumptions.

3. Cost Model

3.1. Model structure

For each source, the total program cost over T years comprises:

$$C_{\text{total}} = K + \sum_{t=1}^T [C_{\text{ops}}(t) + C_{\text{transport}}(t)] \quad (1)$$

70 where K is the upfront capital investment, $C_{\text{ops}}(t)$ is the annual operating cost
 with learning curve effects, and $C_{\text{transport}}(t)$ is the annual cost of transporting
 extracted water to L4/L5.

3.2. Capital costs

Capital costs include extraction equipment (K_{ext}), transport vehicle fleet
 (K_{trans}), anchoring/excavation systems (K_{anch}), water processing and elec-
 75 trolysis (K_{proc}), and for the lunar source, surface base infrastructure (K_{base}).
 Table ?? summarizes baseline values.

Table 1: Capital cost parameters by source (baseline values).

Component	NEA (\$B)	Lunar (\$B)	Phobos (\$B)
Extraction	5.0	8.0	12.0
Transport fleet	10.0	15.0	25.0
Anchoring/excavation	0.5	0.2	0.8
Processing	2.0	3.0	3.0
Surface base	—	5.0	—
Total	17.5	31.2	40.8

3.3. Operating costs with learning

Annual operating costs follow a Wright learning curve applied to cumulative water production:

$$c_{\text{ops}}(Q) = c_{\text{ops},1} \cdot \left(\frac{Q}{Q_0} \right)^b \quad (2)$$

80 where $c_{\text{ops},1}$ is the first-unit operating cost per kg, Q is cumulative production, $Q_0 = 1,000$ kg is the normalization scale, and $b = \ln(\text{LR})/\ln(2)$ with learning rate LR.

3.4. Transport costs

85 Transport cost per kg depends on the delta- v from source to L4/L5, the propulsion system's specific impulse, and the transport vehicle economics:

$$c_{\text{transport}} = c_{\text{base}} \cdot \frac{1}{f_{\text{payload}}} \quad (3)$$

where f_{payload} is the payload mass fraction derived from the Tsiolkovsky equation for the given Δv and I_{sp} .

The key asymmetry between sources is that NEA-to-L4/L5 transfers can use high- I_{sp} electric propulsion ($I_{\text{sp}} \approx 2,500$ s, enabled by magnetically 90 shielded Hall thrusters with demonstrated lifetime $> 3,500$ hours [?]), while lunar surface departure requires chemical propulsion ($I_{\text{sp}} \approx 450$ s). This results in NEA payload fractions of $\sim 35\%$ vs. lunar payload fractions of $\sim 15\%$, a factor of $2.3\times$ advantage for the asteroid source that propagates directly into transport cost.

95 3.5. Production ramp-up

Annual production follows a logistic S-curve from zero to the target production rate:

$$P(t) = P_{\text{target}} \cdot \frac{1}{1 + e^{-k(t-t_0)}} \quad (4)$$

with ramp midpoint t_0 and steepness $k = 2.0$.

3.6. Net present value

100 All costs are discounted at rate $r = 5\%$ (real) to year zero:

$$\text{NPV} = K + \sum_{t=1}^T \frac{C_{\text{ops}}(t) + C_{\text{transport}}(t)}{(1+r)^t} \quad (5)$$

The primary metric is NPV cost per kg of water delivered to L4/L5 over the program lifetime.

4. Monte Carlo Framework

We sample each uncertain parameter from uniform distributions spanning plausible ranges (Table ??), drawing 10,000 independent parameter sets per source.

Table 2: Monte Carlo parameter ranges.

Parameter	Units	NEA range	Lunar range
Water fraction	%	5–15	2–10
Extraction yield	%	50–85	40–75
Energy cost	kWh/kg	1.5–4.0	2.5–6.0
Capital (extraction)	\$B	3–8	5–12
Capital (transport)	\$B	7–15	10–25
Ops cost/kg	\$/kg	100–400	200–600
Δv to L4/L5	km/s	3–7	2–3.5
Transport cost/kg	\$/kg	200–1,000	1,000–4,000
Availability	—	0.70–0.95	0.65–0.90

5. Results

5.1. Baseline deterministic results

Under baseline parameters (Table ??), the NPV cost per kg of water delivered to L4/L5 is:

- **C-type NEA:** \$2,714/kg (30-year total: \$19.5B for 7.2 Mt water)
- **Lunar polar ice:** \$3,755/kg (30-year total: \$43.0B for 11.5 Mt water)
- **Phobos/Deimos:** \$16,342/kg (30-year total: \$43.0B for 2.6 Mt water)

The NEA source is 28% cheaper per kilogram than lunar and 84% cheaper than Phobos/Deimos. Phobos is eliminated from further analysis due to its extreme cost disadvantage.

5.2. Monte Carlo results

Over 10,000 trials, the median NPV cost per kg is:

- **C-type NEA:** \$3,100/kg (P10: \$2,400, P90: \$4,100)
- 120 • **Lunar polar ice:** \$4,700/kg (P10: \$3,600, P90: \$6,200)

The probability that NEA water is cheaper than lunar water is **91.1%** across paired Monte Carlo draws.

5.3. Cost drivers

The NPV cost per kg is dominated by:

- 125 1. **Transport cost** (40–50% of total for NEA, 55–65% for lunar): The EP vs. chemical propulsion asymmetry is the single largest cost driver.
2. **Capital amortization** (30–40%): Higher for NEA in early years due to faster ramp-up; higher for lunar over lifetime due to surface base.
- 130 3. **Operating costs** (15–25%): Learning curve effects reduce this component by 40–60% over 30 years.

5.4. Crossover conditions

Lunar sources become cheaper than NEA under two conditions:

1. **High NEA delta- v** (> 6 km/s): This applies to only $\sim 15\%$ of known C-type NEAs, and target selection can avoid these objects.
- 135 2. **Low NEA water fraction** ($< 4\%$): Inconsistent with OSIRIS-REx data showing CI-chondrite-like composition for C-type NEAs.

Both conditions would need to apply simultaneously to make lunar competitive, a scenario with $< 5\%$ joint probability.

6. Discussion

140 6.1. The transport cost asymmetry

The most robust finding is the structural advantage of electric propulsion for NEA-to-L4/L5 transfers. Magnetically shielded Hall thrusters, now validated through 3,570 hours of wear testing [?] with the AEPS/Gateway program bringing them to TRL 8–9, enable payload fractions $2\text{--}3\times$ higher
145 than chemical propulsion from the lunar surface. This advantage persists

across all plausible parameter ranges and is the primary reason the NEA source dominates.

The lunar source could close this gap with development of electric propulsion for lunar ascent (e.g., from a high lunar orbit staging point), but this adds system complexity and does not address the lower water fraction and higher extraction difficulty.

6.2. *Bennu sample data as ground truth*

The OSIRIS-REx Bennu sample results [? ? ?] provide the first ground-truth calibration for asteroid water extraction models. The CI-chondrite-like composition, with abundant hydrated phyllosilicates and evidence of extensive aqueous alteration, supports the 10% baseline water fraction used in our model. The discovery of evaporite minerals [?] additionally suggests that volatiles may be more accessible than assumed in some earlier models, as evaporites indicate concentrated brine deposits rather than uniformly distributed low-concentration water.

6.3. *Optical mining as an alternative pathway*

Sercel’s Optical Mining concept [?] demonstrated an alternative to mechanical excavation: using concentrated sunlight to thermally extract volatiles without surface contact. This approach inherently manages the thermal volatile preservation challenge (rq-0-40) by controlling the heat input rate and capturing volatiles as they sublimate. Our model is agnostic to extraction method—the key parameters (extraction yield, energy cost, rate) can be set for either mechanical or optical mining.

6.4. *Limitations*

This analysis has several limitations. First, we assume independent parameter distributions; in reality, water fraction and extraction yield may be correlated (higher water content may mean easier extraction). Second, we do not model the multi-trip logistics of asteroid mining in detail—a fleet of transport vehicles must cycle between the NEA and L4/L5, and fleet sizing affects capital cost. Third, our transport cost model uses a simplified payload fraction rather than a full trajectory optimization. Fourth, we do not account for the value of non-water resources co-extracted from asteroids (metals, organics), which could offset costs.

7. Conclusion

180 C-type near-Earth asteroids are the preferred water source for large-scale propellant production at L4/L5 under 91% of plausible parameter combinations, with a median cost advantage of 34% over lunar polar ice. This advantage is driven primarily by the ability to use high- I_{sp} electric propulsion for NEA-to-L4/L5 transfers and by the higher water fraction of CI/CM
185 chondrite material confirmed by OSIRIS-REx sample analysis.

We recommend a development strategy that prioritizes NEA water extraction while maintaining lunar capability as a risk hedge:

1. **Phase 0 (Years 1–5):** Target selection and characterization of 3–5 accessible C-type NEAs using spectroscopy and radar.
- 190 2. **Phase 1 (Years 3–8):** Technology demonstration: thermal extraction of water from asteroid simulant at >100 kg scale in relevant environment.
3. **Phase 2 (Years 6–12):** First operational extraction from a C-type NEA, producing 10–50 tonnes/year.
- 195 4. **Phase 3 (Years 10–30):** Scale to 500+ tonnes/year with fleet expansion.

Lunar capability should be maintained as a parallel track at lower funding level, to be activated if NEA water fraction proves systematically lower than predicted or if specific NEA access constraints emerge.

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